

PROJECT REPORT

LICENSED SOFTWARE REQUEST MANAGEMENT

ServiceNow Service Catalog and Network Request Automation

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Project Documentation

Prepared based on the implemented project and reference report structure.

Technology: ServiceNow

Application: LICENSED SOFTWARE REQUEST MANAGEMENT

Project: Automated Licensed Software Installation Request
Management

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1. INTRODUCTION

1.1 Project Overview

The Licensed Software Installation Request system is a ServiceNow-based solution designed to simplify and automate the process of requesting installation of licensed software. Employees can submit software installation requests through a centralized Service Catalog. The system captures required information such as software name, software version, business justification, required date, employee, target device, manager, department, urgency, and license type.

After submission, ServiceNow creates the request records and processes the request through an automated workflow. The workflow retrieves catalog variables, evaluates the request, validates license information, routes the request for approval, and creates a Software Installation Catalog Task after the required approvals are completed. The task is assigned to the Software Support Team for fulfillment.

High-Level Flow

Employee → Service Catalog → REQ/RITM → License Validation → Approval → Software Asset Approval → Catalog Task → Software Support Team → Installation → Notification → Closure

Proposed Solution — Licensed Software Request Management

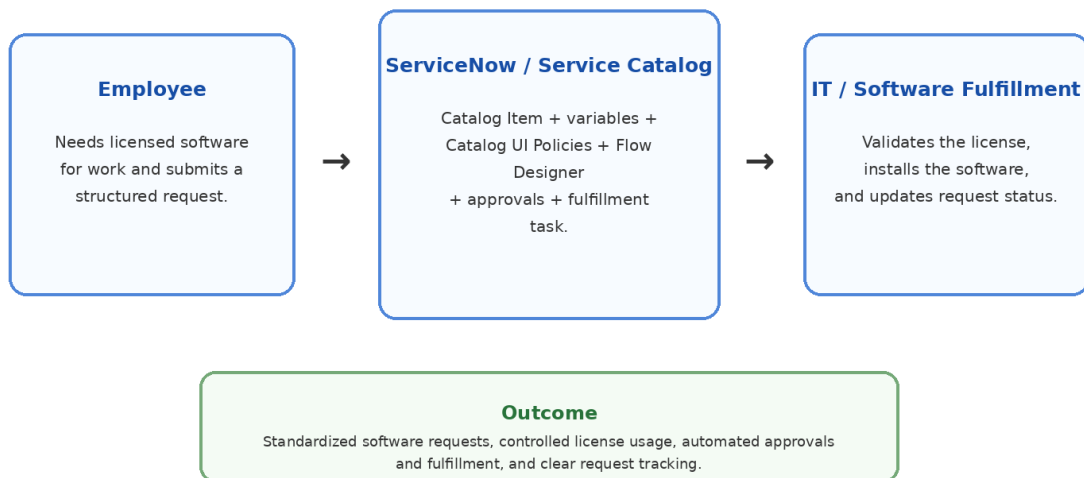


Figure 1: High-level workflow of the Licensed Software Installation Request System.

1.2 Purpose

The purpose of this project is to provide a centralized and structured process for managing licensed software installation requests.

The system aims to:

- Simplify licensed software request submission.
- Capture complete software and employee information.
- Validate license availability before fulfillment.
- Automate approval processing.
- Automatically create software installation tasks.
- Assign tasks to the Software Support Team.

- Improve request tracking and visibility.
- Provide SLA and notification support.
- Maintain controlled access through ServiceNow roles and ACLs.

2. IDEATION PHASE

2.1 Problem Statement

Employees may face difficulties when requesting licensed software because requests can be submitted through unclear or unstructured processes. Incomplete information, missing license checks, manual approvals, and manual task assignment can cause delays in software installation and make request tracking difficult.

Problem Statement Canvas

Parameter	Description
User	Employees requesting licensed software
Problem	Software requests may be incomplete or lack licensing information
Need	A simple and structured request process
Impact	Delays in validation, approval, and installation
How Might We?	How might we provide a centralized system for requesting, validating, approving, and installing licensed software efficiently?

2.2 Empathy Map Canvas

Employee / End User Perspective

Dimension	Employee Perspective
Says	I need a simple way to request the software I require.
Thinks	I want a clear form with relevant software and installation options.
Does	Opens the portal, selects the Licensed Software Installation Request, and submits details.
Feels	Wants fast, reliable, and transparent software support.
Pain Points	Incomplete forms, unclear approval status, license uncertainty, and delayed fulfillment.
Needs / Goals	Simple submission, clear tracking, timely approval, and successful installation.

2.3 Brainstorming & Idea Prioritization

Idea	Priority
Licensed Software Request Catalog Item	High
Catalog Variables	High
Catalog UI Policies	High
Catalog Client Scripts	High
License Validation	High
Approval Process	High
Flow Designer / Workflow Automation	High
Software Installation Task	High
Software Support Team Assignment	High

Notifications	Medium
SLA Monitoring	Medium
Reports and Dashboards	Medium
ACLs and User Roles	High

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The employee journey begins when the user accesses the Service Portal and ends when the licensed software request is fulfilled.

Stage	Employee Action	System Response
Access	Opens Service Portal	Portal displays available services
Discover	Finds Licensed Software Installation Request	Catalog item is displayed
Submit	Enters request details	Variables capture information
Validate	Completes required fields	UI Policies and Client Scripts validate the form
Process	Submits request	REQ and RITM are created
License Check	Waits for validation	Workflow checks configured license information
Approval	Waits for approval	Approval workflow is triggered
Fulfillment	Request is processed	Software Installation Task is created
Completion	Receives completed service	Task and request status are updated

3.2 Solution Requirements

Functional Requirements

ID	Requirement
FR-01	Employees shall access the Licensed Software Installation Request Catalog Item.
FR-02	The system shall capture required software and requester information.
FR-03	The system shall validate mandatory fields.
FR-04	Relevant fields shall be displayed or controlled dynamically.
FR-05	The system shall create REQ and RITM records after submission.
FR-06	The request shall undergo license validation.
FR-07	The request shall be sent for approval.
FR-08	A Software Installation Task shall be created after required approval.
FR-09	The Software Support Team shall process the assigned task.
FR-10	The system shall send relevant notifications and update request status.

Non-Functional Requirements

ID	Requirement
NFR-01	The system should provide a simple user interface.

NFR-02	The system should use ServiceNow access controls.
NFR-03	The workflow should process requests consistently.
NFR-04	The system should provide normal ServiceNow response performance.
NFR-05	The configuration should be maintainable and deployable through an Update Set.

3.3 Logical Data Flow

Employee Input → Catalog Variables → Service Catalog → REQ/RITM → Workflow → License Validation → Approval → Catalog Task → Software Support Team → Task Completion → Notifications / Dashboard

Proposed Solution — Licensed Software Request Management

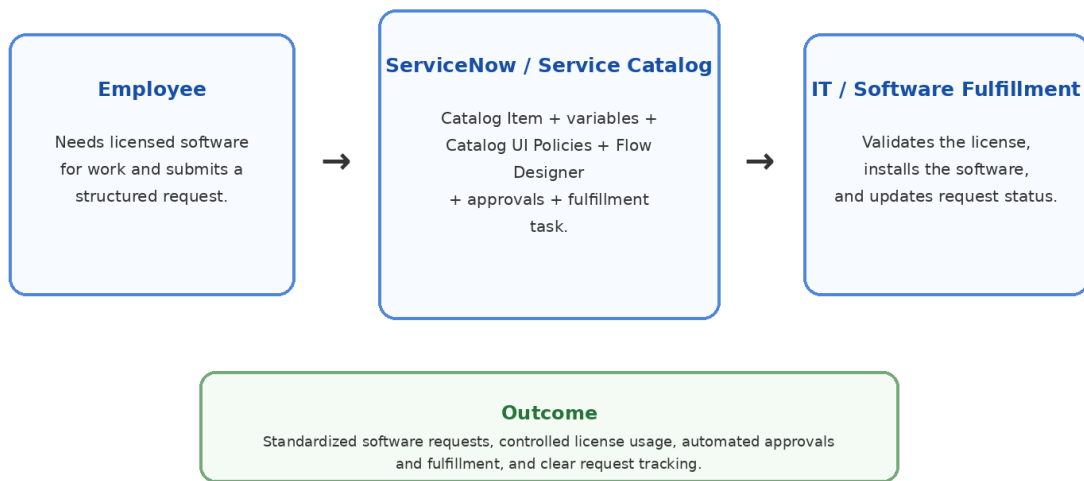


Figure 3: Logical data flow of the Licensed Software Installation Request process.

3.4 Technology Stack

S.No.	Component	Description	Technology
1	User Interface	Employee-facing request interface	Service Portal / Service Catalog
2	Catalog	Licensed software request submission	Service Catalog
3	Variables	Software and installation information	Catalog Variables
4	Rules	Dynamic fields and validation	Catalog UI Policies / Client Scripts
5	Automation	Approval and workflow processing	Flow Designer / Workflow
6	Data	Request and fulfillment records	REQ / RITM / Catalog Tasks
7	References	User and group information	sys_user / sys_user_group
8	Licensing	Software license validation	License / software asset records
9	Communication	Request event messages	ServiceNow Notifications

10	Monitoring	Time-based processing targets	SLA
11	Security	Controlled access	Roles / ACLs
12	Deployment	Configuration packaging	Update Set

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Parameter	Project Fit
Problem	Employees need a simple and structured way to request licensed software.
Solution	A ServiceNow-based Licensed Software Installation Request system.
Customer Benefit	Guided request entry, approval visibility, and predictable fulfillment.
Operational Benefit	Structured requests improve licensing checks, approvals, and installation.
Uniqueness	Combines Service Catalog, UI controls, license validation, approvals, automation, and task fulfillment.
Scalability	Additional software products, approval levels, and deployment integrations can be added later.

4.2 Proposed Solution

The proposed solution is a ServiceNow-based Licensed Software Installation Request system. Employees submit software installation requests through a dedicated Catalog Item. Catalog Variables capture request information, while UI Policies and Catalog Client Scripts support validation and dynamic form behavior.

After submission, the request is processed by SIRA_Software_Installation_Workflow. The workflow retrieves catalog variables, evaluates priority, performs license lookup, routes approval, and proceeds through software asset approval. Once the request is approved and validated, a Software Installation Catalog Task is created and assigned to the Software Support Team.

Notifications communicate approval, assignment, decision, and completion events. SLAs provide time-based monitoring, while the dashboard provides operational visibility.

4.3 Solution Architecture

The solution architecture connects the Service Portal, Licensed Software Request Catalog Item, catalog variables, validation rules, license information, approval process, workflow automation, Software Installation Tasks, notifications, SLAs, security, and dashboard reporting.

Service Portal → Catalog Item → Catalog Variables → REQ/RITM → SIRA Workflow → License Validation → Approval → Asset Approval → Software Installation Task → Software Support Team → Completion → Notifications / Dashboard

Solution Architecture — Licensed Software Request Management

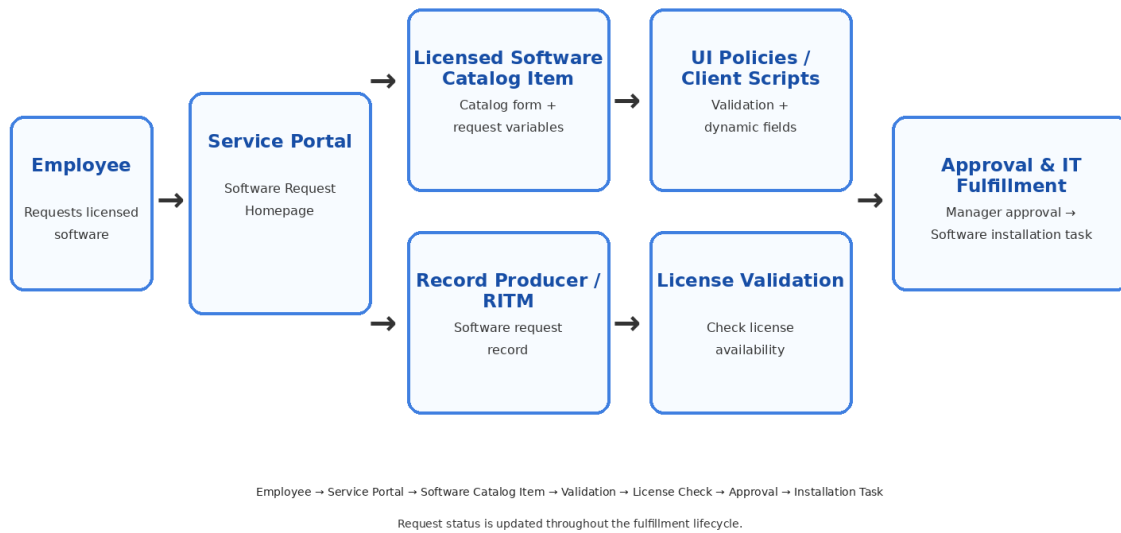


Figure 4: Solution architecture of the Licensed Software Installation Request System.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was divided into six implementation phases to provide a structured development and validation sequence.

Phase	Functional Area	Major Activities	Priority
Phase 1	Requirement Analysis & Planning	Requirements, roles, workflow, approvals, licensing, SLA and security.	High
Phase 2	Service Catalog & Form Configuration	Catalog item, variables, request records and form structure.	High
Phase 3	UI/UX Development & Customization	UI Policies, Catalog Client Scripts and form behavior.	High
Phase 4	Workflow Automation & Approval	Workflow, license validation, approvals and task creation.	High
Phase 5	Testing & Security	Functional testing, UAT, SLA, notification and security validation.	High
Phase 6	Deployment, Documentation & Final Presentation	Update Set, deployment validation, dashboard, documentation and final demonstration.	High

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

Test ID	Scenario	Expected Result	Status
TC-01	Open Service Portal	Portal loads successfully	Pass
TC-02	Find Licensed Software Request	Catalog item is visible	Pass
TC-03	Open request form	Form opens successfully	Pass
TC-04	Enter request details	Variables are available	Pass
TC-05	Validate fields	Mandatory validation works	Pass
TC-06	Submit request	REQ and RITM are created	Pass
TC-07	License validation	Configured license check is executed	Pass
TC-08	Approval process	Request goes for approval	Pass
TC-09	Create Software Installation Task	Task is created after approval	Pass
TC-10	Assignment	Task is assigned to Software Support Team	Pass
TC-11	Notification	Configured notification is generated	Pass
TC-12	End-to-end test	Complete workflow works	Pass

6.2 User Acceptance Testing (UAT)

UAT ID	Scenario	Acceptance Criteria	Result
UAT-01	Open Service Portal	Portal accessible	Accepted
UAT-02	Find Licensed Software Request	Catalog item visible	Accepted
UAT-03	Open form	Request form opens	Accepted
UAT-04	Enter software details	Required options available	Accepted
UAT-05	Validate fields	Required validation works	Accepted
UAT-06	Submit request	Request created	Accepted
UAT-07	License validation	License check works	Accepted
UAT-08	Approval	Approval works	Accepted
UAT-09	Task creation	Task assigned	Accepted
UAT-10	Complete request	Request completed	Accepted

UAT Conclusion

The tested Licensed Software Installation Request workflow satisfies the documented acceptance criteria for the implemented scope.

6.3 Performance Testing

The implemented workflow was tested for normal interactive usage within the ServiceNow development environment.

- Catalog form loading.

- Request submission.
- Catalog variable validation.
- Workflow execution.
- License validation processing.
- Approval workflow execution.
- Software Installation Task creation.
- Notification generation.
- Request completion.

Note: Large-scale concurrent-user load testing was outside the scope of the project.

7. RESULTS

7.1 Output Screenshots

Insert the actual project screenshots in the following locations, matching each caption.

The screenshot displays a web-based form for submitting a software request. The form is titled 'Catalog Item' and 'SCFA00000000'. It includes several sections for data entry:

- Header Section:** Contains fields for 'Number' (SCFA00000000), 'Assigned to' (empty), 'Configuration Item' (empty), and 'Active' (checked). It also has dropdowns for 'Approval' (Approved), 'Priority' (None), and 'State' (Open). There are also fields for 'Request Item' (SCFA00000000) and 'Requested for' (System Administrator).
- Variables Section:** A large section with multiple input fields and dropdowns:
 - 'Which software version you want?' with a dropdown menu.
 - 'Which software you want?' with a dropdown menu.
 - 'Business Justification' with a text area.
 - 'By which date you need it?' with a date picker set to '2024-09-10'.
 - 'Enter employee name' with a text field containing 'David Lee'.
 - 'Where you want to install your software/device?' with a dropdown menu set to 'H00000000'.
 - 'Enter your manager name' with a text field containing 'Bill Richman'.
 - 'Enter your department name' with a text field containing 'Development'.
 - 'Urgency' with a dropdown menu set to 'Medium'.
 - 'Select License Type' with a dropdown menu set to 'LICENSED'.
- Footer Section:** Includes a 'Short description' field with the text 'Install Requested Software', a 'Description' field, and a 'Work notes' field.

The form is styled with a light blue header and a white body. It includes a 'Submit' button at the bottom right.

Fig 7.1 Licensed Software Request Catalog Item / Submitted Request

An SLA starting now will breach on 2026-09-17 11:00:00 (Actual elapsed time: 7 Days 6 Hours 23 Minutes)

Name	Software Request_Normal Priority	Application	Global
Type	SLA	Duration type	User specified duration
Target	--- None ---	* Duration	Days 2 Hours 00 00 00
Table	Requested Item [rc_req_item]	Schedule source	SLA definition
Flow	Default SLA flow	* Schedule	8-5 weekdays excluding holidays
Enable logging	☐	Timezone source	The caller's time zone
Active	☒		

The conditions under which the new SLA will be attached and canceled

Start condition: Add Filter Condition | Add OR Clause

All of these conditions must be met

State	is	Open	AND	OR	X
Urgency	is	2 - Medium	AND	OR	X

Retroactive start: ☐

When to cancel: Cancel conditions are met

Cancel condition: Add Filter Condition | Add OR Clause

-- choose field -- | -- operator -- | -- value --

Update | Delete

Related Links

Show Schedule

Validate SLA Definition

ISN Linked Versions (2)

SLA Breakdown definitions | SLA breakdown def - Search

SLA Definition = Software Request_Normal Priority

SLA breakdown definition

No records to display

Fig 7.2 Configured Catalog Variables

Creating Task: SCTASK00000003

Number: SCTASK00000003

Assigned to: [User]

Configuration item: [Item]

Active: ☒

Approval: Approved

Priority: --- None ---

State: Open

Request item: RTH00000004

Requested for: System Administrator

Variables

* Which software version you want?

Latest Supported Version

* Which software you want?

Adobe Acrobat

* Business Justification

For file compression

* By which date you need it?

2024-09-12

* Enter employee name

David Loo

* Where you want to install your software/device?

PC007-004

Enter your manager name

Brad Robinson

Enter your department name

Development

* Urgency

Medium

* Select License Type

LICENSED

Short description: Install Requested Software

Description:

Work notes: Work notes

Fig 7.3 Submitted Software Installation Request with Variables

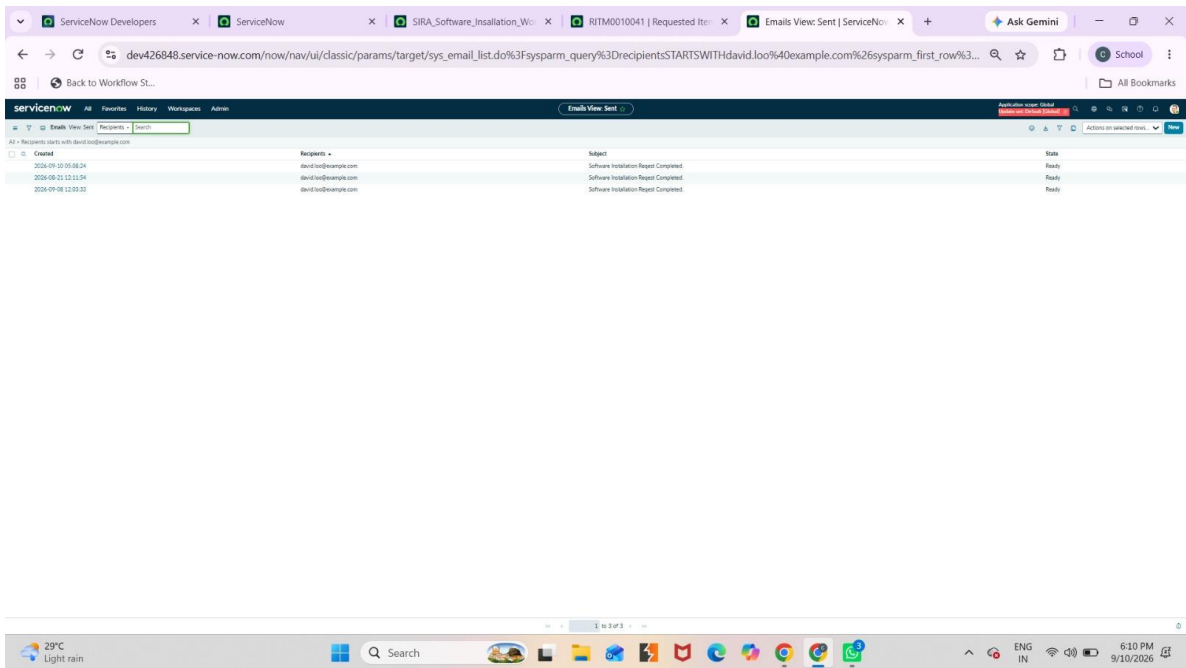


Fig 7.4 User-Friendly Service Catalog Form

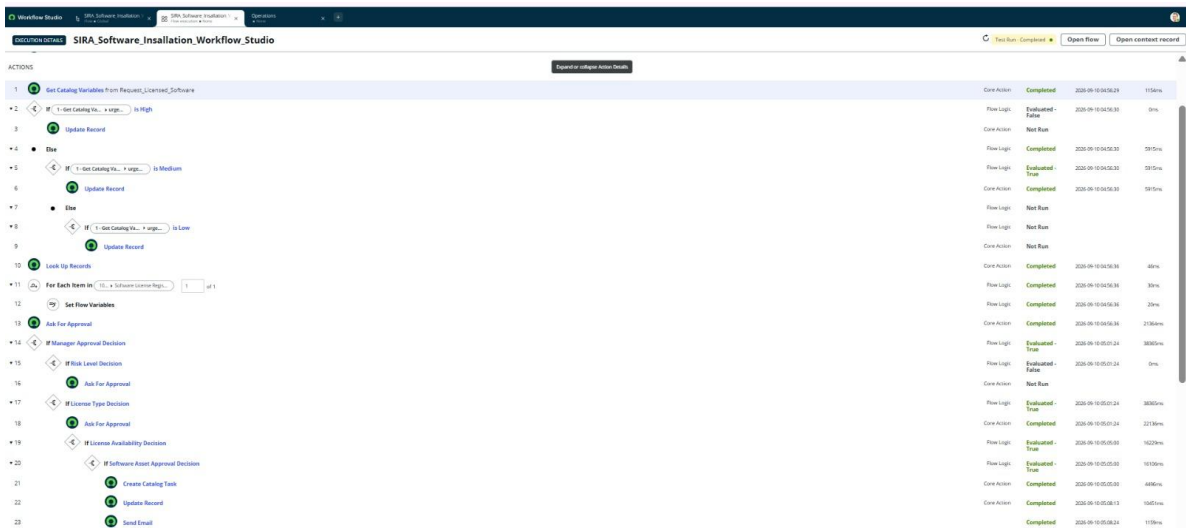


Fig 7.5 SIRA Software Installation Workflow – Workflow Automation Configuration

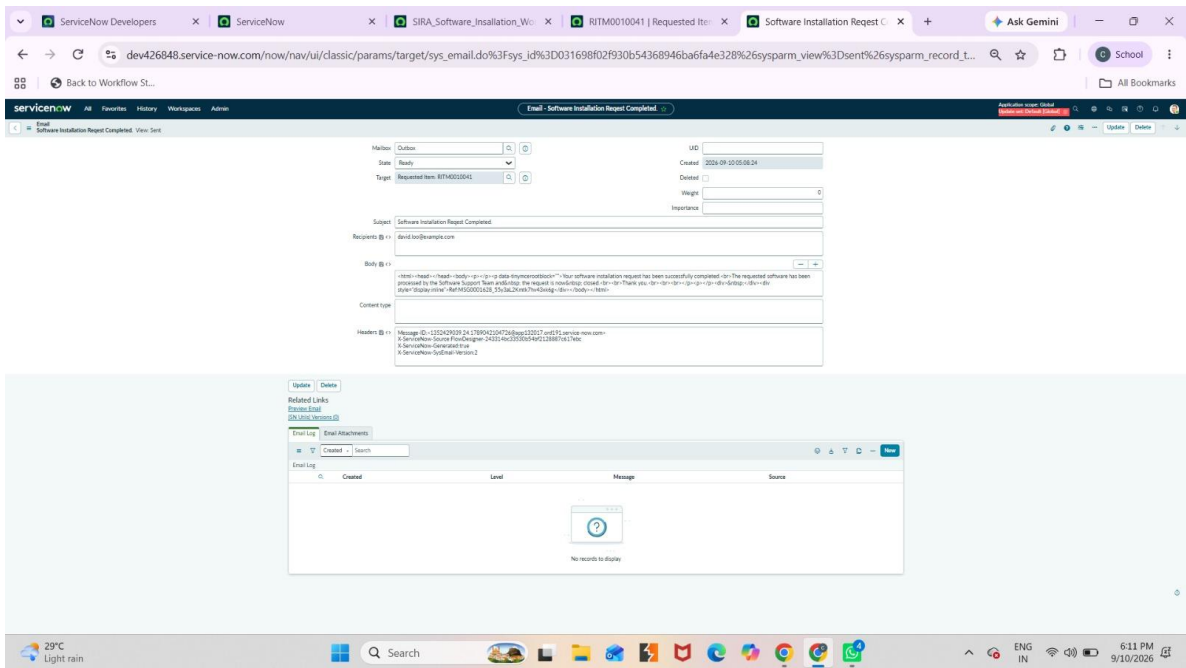


Fig 7.6 Approval Required Notification Configuration

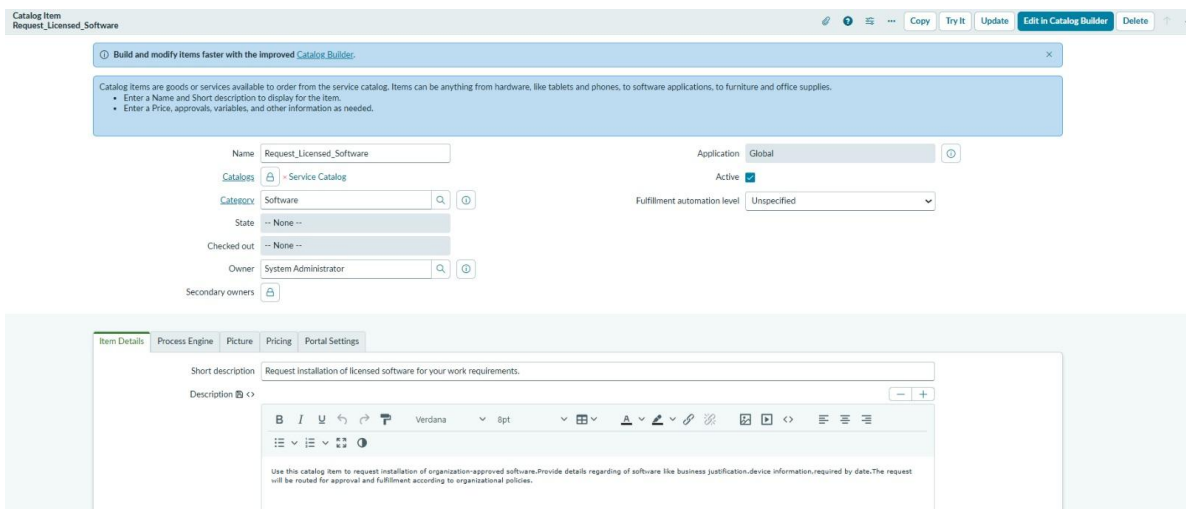


Fig 7.7 Approval / Decision Notification Configuration

Name

Software Request_High Priority

Type

SLA

Target

-- None --

Table

Requested Item [sc_req_item]

Flow

Default SLA flow

Enable logging

☐

Active

☒

Application

Global

Duration type

User specified duration

* Duration

Days 1 Hours 00 00 00

Schedule source

SLA definition

* Schedule

8-5 weekdays excluding holidays

Timezone source

The caller's time zone

Start condition

Pause condition

Stop condition

Reset condition

The conditions under which the new SLA will be attached and canceled

Start condition

Add Filter Condition

Add OR Clause

All of these conditions must be met

State

is

Open

Urgency

is

1 - High

AND

OR

X

AND

OR

X

Retroactive start

☐

When to cancel

Cancel conditions are met

Cancel condition

Add Filter Condition

Add OR Clause

-- choose field --

-- oper --

-- value --

Update

Delete

Related Links

Show Schedule

Validate SLA Definition

ISN Utilist Versions (2)

SLA Breakdown definitions

SLA breakdown defi -

Search

SLA Definition = Software Request_High Priority

SLA breakdown definition

No records to display

Fig 7.8 Software Installation Task / Assignment

Notification - Software Request-Approval Required

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify

- When to send the notification
- Who receives the notification
- What content is in the notification

Name

Software Request-Approval Required

Application

Global

* Table

Approval [sysapproval_approver]

Mandatory

☐

* Category

Uncategorized

Active

☒

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest

☐

When to send

Who will receive

What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the Table specified above
- The specified event is fired
- Via a Flow Action

Send when

Record inserted or updated

Updated

☐

Inserted

☒

Conditions

Add Filter Condition

Add OR Clause

State

is

Requested

AND

OR

X

Preview Notification

Update

Delete

Related Links

Show Notification Scripts

Advanced view

Run Point Scan

ISN Utilist Versions (2)

Fig 7.9 Fulfillment / Completion Evidence

An SLA starting now will breach on 2026-09-17 11:00:00 (Actual elapsed time: 7 Days 6 Hours 23 Minutes)

Name: Software Request_Normal Priority

Type: SLA

Target: -- None --

Table: Requested Item [rc_req_item]

Flow: Default SLA flow

Enable logging: ☐

Active: ☒

Application: Global

Duration type: User specified duration

* Duration: Days 2 Hours 00 00 00

Schedule source: SLA definition

* Schedule: 8-5 weekdays excluding holidays

Timezone source: The caller's time zone

Start condition: Pause condition: Stop condition: Reset condition:

The conditions under which the new SLA will be attached and canceled

Start condition: Add Filter Condition Add OR Clause

All of these conditions must be met

State is Open

Urgency is 2 - Medium

AND OR

Retroactive start: ☐

When to cancel: Cancel conditions are met

Cancel condition: Add Filter Condition Add OR Clause

-- choose field -- -- op -- -- value --

Update Delete

Related Links

Show Schedule

Update SLA Definition

SN Utility Versions (2)

SLA Breakdown definitions: SLA breakdown defi - Search

SLA Definition - Software Request_Normal Priority

SLA breakdown definition

No records to display

Fig 7.10 High Priority SLA / Service Configuration Evidence

Name: Software Request_High Priority

Type: SLA

Target: -- None --

Table: Requested Item [rc_req_item]

Flow: Default SLA flow

Enable logging: ☐

Active: ☒

Application: Global

Duration type: User specified duration

* Duration: Days 1 Hours 00 00 00

Schedule source: SLA definition

* Schedule: 8-5 weekdays excluding holidays

Timezone source: The caller's time zone

Start condition: Pause condition: Stop condition: Reset condition:

The conditions under which the new SLA will be attached and canceled

Start condition: Add Filter Condition Add OR Clause

All of these conditions must be met

State is Open

Urgency is 1 - High

AND OR

Retroactive start: ☐

When to cancel: Cancel conditions are met

Cancel condition: Add Filter Condition Add OR Clause

-- choose field -- -- op -- -- value --

Update Delete

Related Links

Show Schedule

Update SLA Definition

SN Utility Versions (2)

SLA Breakdown definitions: SLA breakdown defi - Search

SLA Definition - Software Request_High Priority

SLA breakdown definition

No records to display

Fig 7.11 Normal Priority SLA / Service Configuration Evidence

Functional & Performance Testing Results — Licensed Software Request

TC-01	Open Service Portal	PASS
TC-02	View Licensed Software Catalog Item	PASS
TC-03	Open Software Request Form	PASS
TC-04	Enter Software Request Details	PASS
TC-05	Select Licensed Software	PASS
TC-06	Validate Mandatory Fields	PASS
TC-07	Submit Software Request	PASS
TC-08	Approval Process	PASS
TC-09	License Validation & Fulfillment Task	PASS
TC-10	End-to-End Installation Request Flow	PASS

10 / 10 functional scenarios passed

All licensed software request management scenarios validated successfully.
Concurrent-user load benchmarking was not performed.

Fig 7.12 Functional and Performance Testing Results

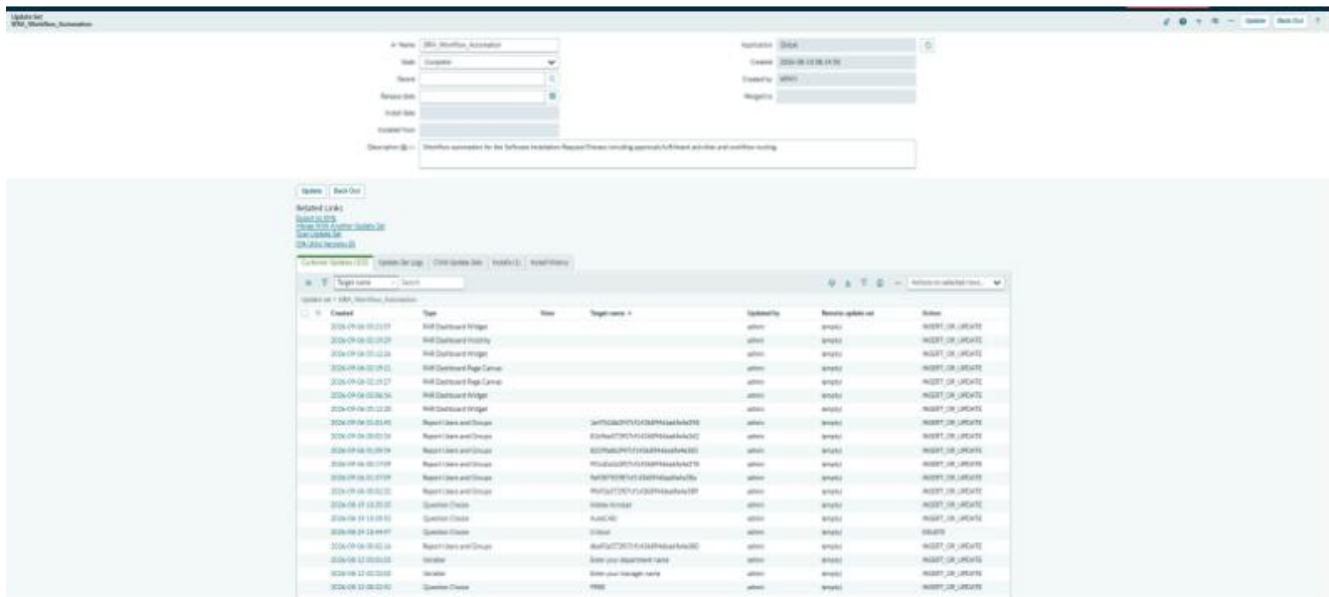


Fig 7.13 Update Set / Deployment Configuration Evidence

7.2 Work Item Status Validation

Component	Status	Validation
Licensed Software Request Catalog	Completed	Catalog item created and accessible
Catalog Variables	Completed	Required fields configured
UI Policies	Completed	Dynamic behavior configured
Catalog Client Scripts	Completed	Validation configured
License Validation	Completed	License check configured
Approval Process	Completed	Approval configured
SIRA Workflow	Completed	Workflow automation configured

Software Installation Task	Completed	Task created for fulfillment
Software Support Team	Completed	Fulfillment assignment configured
Notifications	Completed	Request events communicated
SLA	Completed	Priority-based SLA configured
Service Portal	Completed	Catalog accessible
Security	Completed	Roles and ACL controls configured
Dashboard	Completed	Request/license visibility configured
Update Set	Completed	Deployment package prepared

8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides centralized licensed software request submission.
- Reduces incomplete request information.
- Improves request standardization.
- Automates license validation and approvals.
- Automatically creates fulfillment tasks.
- Improves visibility into the request lifecycle.
- Provides notifications and SLA monitoring.
- Reduces manual effort for the Software Support Team.
- Supports controlled deployment and auditability.

Disadvantages

- Depends on correct ServiceNow configuration.
- Initial configuration requires ServiceNow platform knowledge.
- Complex workflows may require additional maintenance.
- License data must be accurate for reliable validation.
- Advanced deployment integrations may require additional development.

9. CONCLUSION

The Licensed Software Installation Request through the Service Catalog system provides a structured approach to managing licensed software installation requests using ServiceNow.

The solution enables employees to submit requests through a centralized catalog, captures required information, validates request and licensing data, supports approval processing, and automates Software Installation Task creation. Tasks are assigned to the Software Support Team for fulfillment, while notifications, SLAs, dashboards, roles, and ACLs improve operational visibility and control.

The project demonstrates how ServiceNow capabilities such as Service Catalog, Catalog Variables, UI Policies, Catalog Client Scripts, Flow Designer / Workflow, Approval Engine, license validation, Task Management, Notifications, SLA, Security, and Update Sets can be combined to improve operational efficiency and the employee experience.

10. FUTURE SCOPE

The system can be enhanced in the future with:

- Integration with automated software deployment tools.
- Advanced dashboards and reporting.
- Real-time license inventory synchronization.
- Automatic priority calculation.
- Additional approval levels for restricted or high-cost software.
- Automatic license renewal and reclamation.
- Email and mobile notifications.
- AI-based request classification.
- Virtual Agent support.
- Knowledge articles for common software requests.
- Integration with endpoint management platforms.

11. APPENDIX

Appendix A – Technology Components

- ServiceNow Service Catalog
- Catalog Items
- Catalog Variables
- Catalog UI Policies
- Catalog Client Scripts
- Flow Designer / Workflow
- Approval Engine
- REQ and RITM Records
- License / Software Asset Information
- Software Installation Catalog Tasks
- sys_user
- sys_user_group
- Notifications
- SLA
- Roles and ACLs
- Dashboard
- Update Set

Appendix B – Project Flow

Employee Request → Service Catalog → Catalog Variables → REQ/RITM → License Validation → Approval → Software Asset Approval → Software Installation Task → Software Support Team → Installation → Notification → Closure